

CODE SPORT



In this month's column, we discuss the topic of automatic relation extraction from text using natural language processing techniques.

In continuation of our discussion on natural language processing (NLP), let's look at how it can be used in extracting information automatically from text. Given the exponential growth in information over the Web, extracting and organising knowledge manually is no longer feasible. Automated tools are needed to read the ever growing text available on Web, extract facts and information from it, and be able to organise it in a manner amenable to answering questions from users.

Given that search engines are used increasingly for finding answers to questions (and not just general information), it is important that such search engines go beyond merely returning pages of search result links. For instance, if a user types the following query into the search engine today, "Who is the current president of the United States?", it is not user friendly to return thousands of search engine results. Instead, a single-line answer such as, "Barack Obama is the current president of the United States" is the most user friendly and acceptable result possible from the search engine. If you try this query on Google's search engine, it would return you the one-line answer, "Barack Obama", with additional information on him in a small box, and standard search engine results displayed below. Google's ability to answer factual questions directly is powered by means of its Knowledge Graph (<http://googleblog.blogspot.co.uk/2012/05/introducing-knowledge-graph-things-not.html>). This is a knowledge base curated from multiple data sources such as Freebase, Wikipedia and Google's search engine results. Knowledge Graph is a knowledge base. Just as Google uses it to improve its ability to answer factual queries, Microsoft developed a knowledge base known as Satori Knowledge Base to power its Bing search engine results.

A knowledge base consists of unstructured information, represented in a structured form so that it can be queried easily. A knowledge base system typically contains facts represented in a knowledge

base, and an inference engine that can reason using these facts to derive new facts and to detect any inconsistencies among these facts. Knowledge Graph and Satori are knowledge bases harvesting world knowledge from the Web. Knowledge bases can be curated either manually or through automatic means. For Web scale knowledge bases, manual curation is not feasible. Hence, an automatic means of extracting information from the Web is needed to populate such Web-scale knowledge bases.

Let us now discuss how NLP techniques can be used to extract information automatically from text documents to populate knowledge bases. This task is popularly known as Information Extraction (IE). It is the task of processing human language texts by NLP techniques. The core functionality of IE is concerned with extracting information present in unstructured/semi-structured machine readable texts and representing it in a structured manner. Extracting information about entities and their relationships is the core task of IE.

Information extraction can be done specific to a particular domain or it can be done in a domain-independent manner. Since unstructured information needs to be represented and organised in a structured manner, IE requires that a structure capable of supporting queries on the extracted information needs to be defined first. This structure defines the type of entities and the relationship between them. For example, let us consider a domain-specific information extraction task. You have been given a set of medical research texts and are asked to extract information pertaining to various diseases. The entities related to this information extraction can potentially be the disease names, their associated symptoms, the drugs used to treat the different diseases, etc. Some of the possible relations include: 'Disease X has symptom Y', 'Disease X is treated by Drug Z', etc. These textual relationships are defined in the domain ontology by ontological relations such as TREATS and 'Has